

Manav Kumar

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EDUCATION

University of California, Berkeley College of Computing, Data Science, & Society

May 2026 | GPA: 3.86/4.0

Bachelor of Arts in Data Science & Bachelor of Arts in Statistics

Relevant Coursework: Data Science Techniques, Data Ethics, Data Structures, Computer Programs, Linear Algebra

Organizations: Data Science Society at Berkeley, Delta Sigma Phi, Sustainable Entrepreneurship at Berkeley

Awards & Honors: GALLO Scholarship (6% acceptance), Northern California Scholarship Federation Award (1% acceptance)

PROFESSIONAL EXPERIENCE

Machine Learning & Data Analysis Intern Rotation 1 | Modesto, CA

June 2023 – August 2023

E. & J. Gallo Winery

- Created AWS Redshift to SageMaker pipeline to manage customer wine purchase data from Fetch Rewards and perform EDA
- Curated ML lifecycle Proof of Concept (POC) and leveraged SageMaker to test the process with customer wine purchase data
- Presented findings to 20+ senior statisticians, leading to the adoption of my POC in the company's restructuring towards ML

Machine Learning & Data Analysis Intern Rotation 2 | Modesto, CA

June 2024 – August 2024

E. & J. Gallo Winery

- Adapted the ML Lifecycle and POC by cleaning and processing Fetch Rewards data via Time Series Analysis using PySpark
- Predicted customer churn rate with up to 70% accuracy using both statistical models and multiple Machine Learning models
- Optimized feature selection and engineering, through standardization, which increased the accuracy up to 80% per ML model

Lead Data Analyst | Modesto, CA

July 2022 – August 2023

Parks Youth Committee – Tuolumne River Trust

- Audited 6 parks using ArcGIS with 300+ data points for accessibility and quality to highlight disparities between parks
- Advocated for increased funding for low-income parks, leading to long-term infrastructure redesigns in the city of Modesto
- Taught the next cohort of youth leaders how to use ArcGIS, scrape data from the web, and perform EDA on data gathered

RESEARCH & PROJECTS

Undergraduate Researcher | Berkeley, CA

June 2024 – Present

UC Berkeley Center for Targeted Machine Learning

- Analyzed billions of medical records using SQL and PySpark to assess drug impact on Long COVID diagnosis and symptoms
- Used Targeted Learning to train a Superlearner ML algorithm to determine causation between drug and COVID interactions
- Onboarded other research assistants, taught them how to use our databases and perform SQL operations for research analysis

Undergraduate Researcher | Berkeley, CA

January 2025 – Present

UC Berkeley Center for the Science of Psychedelics

- Created a Logistic Regression model to identify cognitive effect sentences from trip reports scraped from Erowid's online data
- Clustered sentences using HDB-Scan to hypothesize which psychedelic compounds correlated with cognitive sentence type
- Performed hypercluster analysis to highlight the macro and micro cognitive effects patterns across different substances

Personal Project | "RAG from Scratch" | Berkeley, CA

January 2025 – Present

- Extracted online Q & A prompts on Reddit about UC Berkeley using Reddit's API to scrape post, comment, and upvote data
- Used Hugging Face's online model database to implement GPT-2 tokenization for training a RAG Model created from scratch
- Fine-tuned the retriever and generator components to generate contextually relevant answers based on the prompt input

Personal Project | "AirBnb Analysis" | Berkeley, CA

January 2025 – Present

- Led a 4 person team in performing EDA and creating visuals to map out and learn from AirBnb data in Seattle to predict price.
- Trained a forest regression model with up to 85% accuracy to input multiple AirbnB features and output a price prediction.
- Presented findings to Data 198, a course used to teach students how the data science lifecycle works with actionable insights.

SKILLS & INTERESTS

Skills: AWS, Machine Learning, JupyterHub, NLP, Git/GitHub, Data Science Lifecycle, Research, Communication, Presenting

Languages & Libraries: Java, SQL, DBeaver, Python, R/R-Studio, Pandas, NumPy, Seaborn, Matplotlib, PySpark, Scikit-Learn

Interests: Gym, Self-Improvement, Philosophy, Space, Reading, Teaching, Personal Finance, Video Games, Podcasts, Hiking